Practical Predefined-Time Output-Feedback Consensus Tracking Control for Multiagent Systems

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***Abstract*—This article addresses the issue of output-feedback consensus control of multiagent systems under the directed topology and subject to bounded external disturbances. By employing a smooth time-varying function, a distributed practical predefined-time (PPT) observer is developed to estimate the reference trajectory for the entire team (i.e., the leader's state) and a practical preset-time extended-state observer is also proposed to estimate bounded disturbances and unmeasurable system states. Next, a novel continuous and nonsingular PPT consensus control law is designed on the basis of the observers. Furthermore, the designed control protocol can achieve PPT stability, that is, consensus tracking errors are enforced to a neighborhood around zero within a predetermined time, which can be specified *a priori*, independent of initial states of agents and/or any other design parameters. Finally, illustrative numerical examples, including a comparative one, are provided to demonstrate the performance of the present predefined-time control approach.**

***Index Terms*—Consensus control, multiagent systems (MASs), practical predefined-time (PPT) stability, sliding-mode control.**

I. Introduction

COOPERATIVE control of multiagent systems (MASs) has received widespread attention in recent years from many scientific communities because of its comprehensive applications in spacecraft formations [1], [2], [3], [4]; intelligent robots [5], [6]; unmanned aerial vehicles [7], [8]; and so on. Consensus control, which is a primary problem of cooperative control, refers to a group of agents that can achieve an agreement on specific states or outputs.

Currently, the majority of consensus control strategies for MASs are developed to enforce system states to reach an equilibrium asymptotically. Contrary to the asymptotically stable control law, finite-time control schemes have advantages of higher accuracy control performance and better disturbance resistance ability [9], [10]. The existing approaches for the derivation of finite-time consensus control include the terminal sliding-mode control method [11], [12]; the "adding a power integrator" technique [10], [13], [14]; and the homogeneous method [15]. However, the finite-time control has one shortcoming; that is, the convergence time is dependent on initial conditions. Hence, fixed-time control algorithms that can produce requested control accuracy in a determined time without relying on initial conditions are proposed to overcome such a problem [16]. In the literature, various studies on fixed-time consensus control have been carried out [17], [18], [19], [20], [21], [22], [23].

Although the fixed-time controller has an enormous advantage over the finite-time control law, the exact time of convergence might be hard to compute. The predefined-time control protocol [24], [25], [26], [27], [28], [29], an advanced notion of stability, implies that the settling time can be specified in advance by the controller designer. In [25], [26], [27], [28], and [29], the sliding-mode control was used to design predefined-time control schemes for second-order nonlinear systems. However, the predefined-time stability could not be achieved during the sliding phase in [25] due to the use of a linear sliding-mode manifold and there is a singularity problem in implementing the predefined-time controllers proposed in [26], [27], and [28] because of a term with a negative power in the designed sliding-mode surface. In [30], the time-varying technique was applied to design a predefined-time output-feedback controller for second-order nonlinear systems; however, external disturbances were not taken into account. Furthermore, the control schemes presented in [24], [25], [26], [27], [28], [29], and [30] were designed for a single agent system rather than for a team of agent systems, and the extension of these predefined-time controllers to consensus control of a group of agent systems is not straightforward because of coupling among neighboring agents.

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Recently, the issue of predetermined-time consensus control for MASs has been addressed in some studies by applying the terminal sliding-mode control method [31] and the time-varying function technique [32], [33], [34], [35], [36], [37]. Aided by the terminal sliding-mode control approach, a predefined-time consensus tracking control scheme was developed for double-integrator MASs with undirected communication topology [31]. Based on the time-varying function technique, predefined-time consensus controllers were designed for the first-order [32], [33], [34], second-order [35], and high-order MASs [36], [37]. However, the time-varying function used in [32], [33], [36], and [37] would grow unbounded toward the prescribed time, which might lead to a singularity problem in implementing the designed controllers. In addition, external disturbances were not considered in [31], [32], [33], [34], [36], and [37]. Although external disturbances have been taken into account in [35], the designed consensus tracking control law is discontinuous, which may cause chattering. However, chattering is usually not desirable in practical situations since it may excite high-frequency unmodeled system dynamics. Furthermore, as the control inputs are requested to exchange between neighboring agents in [35], the communication burden may increase, resulting in a communication loop problem in the implementation of the controller.

In many practical applications, concerns on sensor failures or cost cutting in onboard sensors, some states may not be measurable for the controller derivation. However, few studies have addressed the issue of distributed fixed/predefined-time output-feedback consensus tracking control of a group of agent systems in the absence of some state measurements. In [23], by using the fixed-time observers, the homogeneous theory, and the homogeneous Lyapunov function approach, a distributed fixed-time output-feedback consensus tracking control scheme was put forward for the case when the communication graph is directed and the leader's information is available only to a subset of the team members. However, the exact convergent time of the resultant closed-loop system is difficult to compute, and the magnitude of the applied control inputs might be very large when the system states are far away from the equilibrium as there is a term with a power greater than one in the designed control scheme. Based on the time-varying function technique, Li *et al.* [37] proposed a preset-time consensus control algorithm for high-order MASs under a directed graph. But there may exist a singularity problem in implementing the controller given that the time-varying function will become unbounded toward the preset time, and external disturbances are not taken into account.

In this article, we consider the issue of practical predefined-time (PPT) output-feedback consensus tracking control for second-order nonlinear MASs under the directed topology and in the existence of bounded external disturbances when the leader's information is available only to some members of the group. The controller derivation becomes much more challenging because the PPT consensus, partial-state measurements, the presence of bounded external disturbances, and the asymmetric property of directed graphs are simultaneously involved. Thus, based on the smooth time-varying function technique, the distributed PPT state observer, the PPT extended-state observer, and the sliding-mode control method, a continuous and nonsingular PPT output-feedback consensus tracking controller is proposed. The major contributions of this article are twofold. First, new distributed PPT state observers and PPT extended-state observers are proposed by employing a smooth time-varying scaling function. Second, a novel distributed PPT output-feedback consensus tracking control law is developed for second-order MASs under directed graphs and in the existence of bounded external disturbances. Unlike the time-varying function-based predetermined-time consensus controllers in [30], [32], [33], [34], [35], [36], and [37], the time-varying function used in the present paper is C^∞ on $(0, +\infty)$. When compared with the predetermined-time output-feedback consensus controller reported in [37], there is no singularity problem in implementing the proposed controller, and external disturbances are explicitly taken into consideration. Compared with the sliding-mode-control-based predefined-time controllers in previous studies [25], [26], [27], [28], the PPT control scheme derived here can guarantee the PPT stability in both the reaching phase and the sliding phase without suffering from any singularity problem. In contrast to the distributed fixed-time output-feedback controller in [23], the proposed control protocol can predefine the settling time without dependence on any other parameters and exhibit additional advantages in reducing the magnitude of the control inputs and saving energy.

The remainder of this article is organized as follows. Some preliminaries and the problem statement are provided in Section II. A distributed PPT output-feedback consensus tracking control scheme is designed in Section III. Illustrative numerical simulations are given in Section IV and conclusions are presented in Section V.

II. Preliminaries and Problem Statement

A. Notations

$\mathcal{N}$ represents $\mathcal{N} = \{1, 2, \ldots, n\}$. col$(\cdot)$ is defined as col$(q_1, q_2, \ldots, q_n) = [q_1^T, q_2^T, \ldots, q_n^T]^T$ for $q_i \in R^m, i \in \mathcal{N}$. $\|\cdot\|$ denotes the Euclidean norm of a vector or the induced norm of a matrix. $\otimes$ refers to the Kronecker product. I_m represents the $m \times m$ identity matrix. $\lambda_{\min}(\cdot)$ and $\lambda_{\max}(\cdot)$ denote the minimum eigenvalue and the maximum eigenvalue of a matrix, respectively.

B. Definitions and Lemmas

Consider the system described by

$$\dot{x} = f(x, t), x(0) = x_0, f(0, t) = 0, x \in R^m \tag{1}$$

where $f : R^m \times R^+ \to R^m$ is a continuous nonlinear function. Let the solution of system (1) be $x(t; x_0)$.

Definition 1: If there exists a preset constant $t_f > 0$ such that

$$x(t; x_0) = 0 \quad \forall t \geq t_f \tag{2}$$

the origin of system (1) is said to be predefined-time stable.

Definition 2: If there exist some $\epsilon > 0$ and a predefined time $t_f > 0$ such that

$$\|x(t; x_0)\| \le \epsilon \quad \forall t \ge t_f \tag{3}$$

the origin of system (1) is said to be PPT stable. The parameter ϵ can be adjusted to the desired level.

Consider a smooth function

$$\varsigma_0(x) = \begin{cases} 0, & (-\infty, 0] \\ e^{-1/x}, & (0, +\infty) \end{cases}, \quad x \in R \tag{4}$$

and define

$$\varsigma_a(t, t_f) = \frac{\varsigma_0(t_f - t)}{\varsigma_0(t_f - t) + \varsigma_0(t)} \tag{5}$$

where $t_f > 0$ is a parameter determined by the controller designer. A time-dependent scaling function $\varsigma(t, t_f, \epsilon) \in R$ is constructed as

$$\varsigma(t, t_f, \epsilon) = \frac{1}{\varsigma_a(t, t_f) + \epsilon} = \frac{\varsigma_0(t_f - t) + \varsigma_0(t)}{(1+\epsilon)\varsigma_0(t_f - t) + \epsilon\varsigma_0(t)} \tag{6}$$

with $0 < \epsilon \ll 1$. The following properties hold for the function $\varsigma(t, t_f, \epsilon)$ defined in (6).

1) $t = 0, \varsigma(t, t_f, \epsilon) = (1/1+\epsilon)$; $t \ge t_f, \varsigma(t, t_f, \epsilon) = (1/\epsilon)$.
2) $\varsigma(t, t_f, \epsilon)$ is C^∞ on $(0, +\infty)$.
3) $\varsigma(t, t_f, \epsilon)$ is monotonically increasing on $(0, t_f)$.

The proof for these three properties is given in the Appendix.

Lemma 1: If there exits a Lyapunov function $V(t)$ which satisfies

$$\dot{V}(t) = -\left(k + \frac{\dot{\varsigma}(t, t_f, \epsilon)}{\varsigma(t, t_f, \epsilon)}\right)V, V_0 = V(0) \tag{7}$$

where $k > 0$ is some constant and $\varsigma(t, t_f, \epsilon)$ is given in (6), then $V(t)$ converges to the region $V \le \epsilon V_0$ with a predesignated time t_f and then approaches to zero asymptotically.

Proof: Defining $\bar{V}(t) = \varsigma V$ and taking the time derivative of $\bar{V}$ result in

$$\dot{\bar{V}} = \dot{\varsigma}V + \varsigma\dot{V} = -k\bar{V}. \tag{8}$$

It follows from (8) that:

$$\bar{V}(t) = \varsigma(0, t_f, \epsilon)V_0 e^{-kt} = \frac{1}{1+\epsilon}V_0 e^{-kt}. \tag{9}$$

As $\bar{V} = \varsigma V$, it can be verified that

$$V(t) = \frac{1}{\varsigma(1+\epsilon)}V_0 e^{-kt}. \tag{10}$$

Hence, we obtain that $V(t)$ approaches to $(\epsilon/1+\epsilon)V_0 e^{-kt_f} \le \epsilon V_0$ when $t \to t_f$ and then converges to zero as $t \to \infty$. ■

Now, we have the following corollary.

Corollary 1: If there is a Lyapunov function $V(t)$ which satisfies

$$\dot{V}(t) = -\left(k + \frac{\dot{\varsigma}(t, t_f, \epsilon)}{\varsigma(t, t_f, \epsilon)}\right)V + \varrho, V_0 = V(0) \tag{11}$$

where $k > 0$ is some constant, ϱ is a small positive constant, and $\varsigma(t, t_f, \epsilon)$ is given in (6), then $V(t)$ will converge to the region $V \le \epsilon V_0 + \varrho/k$ with a predefined time t_f.

Proof: Letting $\bar{V}(t) = \varsigma V$ and calculating the time derivative of $\bar{V}$ lead to

$$\begin{aligned} \dot{\bar{V}} &= \dot{\varsigma}V + \varsigma\dot{V} = -k\bar{V} + \varsigma\varrho \\ &\le -k\bar{V} + \frac{\varrho}{\epsilon} \end{aligned} \tag{12}$$

which results in

$$\begin{aligned} \bar{V}(t) &\le \frac{\varrho}{k\epsilon} + \left(\bar{V}(0) - \frac{\varrho}{k\epsilon}\right)e^{-kt} \\ &\le \frac{\varrho}{k\epsilon} + \varsigma(0, t_f, \epsilon)V_0 e^{-kt} = \frac{\varrho}{k\epsilon} + \frac{1}{1+\epsilon}V_0 e^{-kt}. \end{aligned} \tag{13}$$

Noting that $\bar{V} = \varsigma V$, we have

$$V(t) \le \frac{\varrho}{k\varsigma\epsilon} + \frac{1}{\varsigma(1+\epsilon)}V_0 e^{-kt}. \tag{14}$$

Thus, we have that $V(t)$ approaches to the region $V(t) \le \epsilon V_0 + \varrho/k$ as $t \to t_f$ and then reaches to the region $V(t) \le \varrho/k$ as $t \to \infty$. ■

Remark 1: If the parameter ϵ is taken as $\epsilon = 0$, the time-varying function $\varsigma(t, t_f, \epsilon)$ becomes $\varsigma(t, t_f, 0) = 1/\varsigma_a(t, t_f)$, which implies that $\varsigma(t, t_f, 0) \to \infty$ and $1/\varsigma(t, t_f, 0) = \varsigma_a(t, t_f) \to 0$ as $t \to t_f$, and we can verify that $\dot{\varsigma}/\varsigma \to \infty$ as $t \to t_f$. As $V(t) = V_0 e^{-kt}/(\varsigma(1+\epsilon))$, we can conclude that $V(t) \to 0$ as $t \to t_f$ when $\epsilon = 0$. In this case, similar to the time-varying function used in [32], [33], [36], and [37], $\varsigma(t, t_f, 0)$ will grow unbounded toward the preset time. In the present method, by using a term $\epsilon > 0$ in $\varsigma(t, t_f, \epsilon)$, it can be verified that

$$\dot{\varsigma}(t, t_f, \epsilon) = \frac{\dot{\varsigma}_0(t)\varsigma_0(t_f - t) - \dot{\varsigma}_0(t_f - t)\varsigma_0(t)}{\left[(1+\epsilon)\varsigma_0(t_f - t) + \epsilon\varsigma_0(t)\right]^2}$$

and we can obtain that

$$\frac{\dot{\varsigma}(t, t_f, \epsilon)}{\varsigma(t, t_f, \epsilon)} = \frac{\dot{\varsigma}_0(t)\varsigma_0(t_f - t) - \dot{\varsigma}_0(t_f - t)\varsigma_0(t)}{\left[(1+\epsilon)\varsigma_0(t_f - t) + \epsilon\varsigma_0(t)\right]\left[\varsigma_0(t_f - t) + \varsigma_0(t)\right]}$$

which indicates that the term $\dot{\varsigma}/\varsigma$ is well defined and remains bounded for all $t \ge 0$.

Remark 2: If we choose the parameter ϵ as $\epsilon = \epsilon_0/\epsilon_1$ with $\epsilon_0 > 0$ and $\epsilon_1 > V_0$, then we can prove that $V(t)$ approaches to $V \le \epsilon_0$ within a predetermined time t_f, which indicates that $V(t)$ reaches to a predesignated region around the origin within a preset time.

C. Graph Theory

Consider $n+1$ agents with one (virtual) leader labeled by 0 and n followers denoted by $i \in \mathcal{N}$. The communication topology between the follower agents is described by a directed graph $G = (\Psi, E, A)$, where $A = [a_{ij}] \in R^{n\times n}$ with $a_{ij} \ge 0$ is the adjacency matrix of the weights of G such that $a_{ij} > 0$ if $(\psi_j, \psi_i) \in E$ and $a_{ij} = 0$ otherwise, $\Psi = \{\psi_1, \psi_2, \ldots, \psi_n\}$ is the set of nodes with ψ_i $(i \in \mathcal{N})$ representing the ith agent in the group, and $E \subseteq \Psi \times \Psi$ denotes the set of edges. In addition, $D = \text{diag}\{d_1, d_2, \ldots, d_n\}$ with $d_i = \sum_{j=1}^n a_{ij} (i \in \mathcal{N})$ is the degree matrix and $L = D - A = [l_{ij}]$ is the Laplacian matrix L of G.

The communication topology among the n follower agents and the leader is denoted as $\bar{G}$ with the adjacency matrix $A_0 = \text{diag}\{a_{10}, a_{20}, \ldots, a_{n0}\}$, where $a_{i0} > 0 (i \in \mathcal{N})$ if the ith follower can gain access to information from the leader, otherwise $a_{i0} = 0$. The following assumption about the topology $\bar{G}$ is supposed to hold throughout this article.

Assumption 1: The topology $\bar{G}$ contains a spanning tree in which the leader acts as the root.

Assumption 1 is common in the existing literature on cooperative control of MASs under directed graphs, such as [23], [34], [36], and [38], which can guarantee that all follower agents can obtain access to the leader's information directly or indirectly.

Theorem 1 [38]: Under Assumption 1, there exists a diagonal matrix $Q \in R^{n\times n} > 0$ such that

$$N = QM + M^T Q > 0 \tag{15}$$

with $M = L + A_0$.

D. Problem Statement

The dynamic equations for n follower agents are described by

$$\begin{cases} \dot{x}_{1i} = x_{2i} \\ \dot{x}_{2i} = f_i(x_{1i}, x_{2i}) + g_i(x_{1i})u_i + \vartheta_i \\ y_i = x_{1i} \end{cases} \tag{16}$$

where $x_{1i} \in R^m$ and $x_{2i} \in R^m$ ($i \in \mathcal{N}$) are the system states, $u_i \in R^m$ is the system input, $y_i \in R^m$ is the system output, $f_i(x_{1i}, x_{2i}) \in R^m$ is a continuous nonlinear function with $f_i(0, 0) = 0$, $g_i(x_{1i}) \in R^{m\times m}$ is a nonsingular function, and $\vartheta_i \in R^m$ is the unknown disturbance. The dynamics of the (virtual) leader is

$$\ddot{x}_0 = u_0 \tag{17}$$

where $x_0 \in R^m$ and $u_0 \in R^m$ denote the state and input of the leader.

The following assumptions hold throughout this article.

Assumption 2: The leader's state and its first two derivatives, that is, $x_0, \dot{x}_0$, and $\ddot{x}_0$ (i.e., u_0) are uniformly bounded.

Assumption 3: The disturbance ϑ_i and its first derivative $\dot{\vartheta}_i$ are uniformly bounded so that $\|\dot{\vartheta}_i\| \le \vartheta_M$ ($i \in \mathcal{N}$) with ϑ_M representing some positive constant.

Assumption 4: The nonlinear function $f_i(x_{1i}, x_{2i})$ is assumed to be globally Lipschitz so that $\|f_i(x_{1i}, x_{2i}) - f_i(y_{1i}, y_{2i})\| \le c\|x_i - y_i\|$ ($i \in \mathcal{N}$), where $c > 0$ is a constant, $x_{1i} \in R^m, x_{2i} \in R^m, y_{1i} \in R^m, y_{2i} \in R^m$, $x_i = \mathrm{col}(x_{1i}, x_{2i})$, and $y_i = \mathrm{col}(y_{1i}, y_{2i})$.

Assumption 2 indicates that we can derive a control scheme for $u_i (i \in \mathcal{N})$ so that the agent systems can track a time-varying reference trajectory. Assumption 3 is usually used for the design of an extended-state observer to obtain an estimate of external disturbances [39]. Assumption 4 implies that the nonlinear function in the agent systems [i.e., $f_i(x_{1i}, x_{2i})$] satisfies the Lipschitz growth condition. Many practical systems obey this condition, such as robotic manipulators, inverted pendulum systems, etc.

Control Objective: Design a distributed PPT output-feedback controller for u_i ($i \in \mathcal{N}$) such that

$$\|x_{1i}(t) - x_0(t)\| \le \epsilon \ \forall t \ge t_f \tag{18}$$

for some $\epsilon > 0$ and a predefined time $t_f > 0$, even when only a subset of the follower agents can obtain access to the leader's information and there exist bounded external disturbances imposing on the agent dynamics.

Remark 3: In this article, the (virtual) leader plays a pilot role for follower agents and it is modeled by double-integrator dynamics which is also employed in the existing works, such as [31] and [35]. In practice, the mass-force model meets the double-integrator dynamics and many actual systems (e.g., robotic manipulators [31], wheeled mobile robots [40], etc.) can be transformed into a double-integrator dynamic model through feedback linearization.

III. Main Results

In this section, we consider the issue of output-feedback consensus tracking control for MASs under directed graphs and subject to bounded external disturbances. First, with the help of the property of directed graphs and the time-varying function, a distributed PPT state observer is developed to estimate the leader's states. If each follower can obtain the estimation of the leader's states, the problem of consensus tracking control for MASs will be converted to the problem of tracking control for a single agent system. Second, a time-varying function-based PPT extended-state observer is designed for each follower to estimate unmeasurable system states and bounded external disturbances as well. Finally, by using the time-dependent function technique, the distributed PPT state observer, the PPT extended-state observer, and the sliding-mode control approach, a continuous and nonsingular PPT output-feedback consensus tracking controller is proposed.

A. Design of Distributed PPT Observers

Since only some of the follower agents can receive information from the leader, we propose distributed PPT observers to estimate x_0 and $\dot{x}_0$. The distributed PPT observer is given as follows:

$$\begin{cases} \dot{p}_{1i} = p_{2i} - (\rho_1\theta_1 + \rho_2)\xi_{1i} \\ \dot{p}_{2i} = -(\rho_3\theta_1 + \rho_4)\xi_{2i} \end{cases} \tag{19}$$

where $i \in \mathcal{N}$, $\xi_{1i} = \sum_{j=1}^n a_{ij}(p_{1i} - p_{1j}) + a_{i0}(p_{1i} - x_0)$, $\xi_{2i} = \sum_{j=1}^n a_{ij}(p_{2i} - p_{2j}) + a_{i0}(p_{2i} - \dot{x}_0)$, $\rho_k > 0 (k = 1, 2, 3, 4)$, and $\theta_1 = \rho + \dot{\varsigma}_1/\varsigma_1$ with $\varsigma_1 = \varsigma(t, t_{f1}, \epsilon_1)$. Here, $\rho > 0$, $t_{f1} > 0$ and $\epsilon_1 > 0$ are some constants.

Denote the observer errors by $\tilde{p}_{1i} = p_{1i} - x_0$ and $\tilde{p}_{2i} = p_{2i} - \dot{x}_0 (i \in \mathcal{N})$. Then, it can be obtained that $\xi_{1i} = \sum_{j=1}^n l_{ij}\tilde{p}_{1j} + a_{i0}\tilde{p}_{1i}$ and $\xi_{2i} = \sum_{j=1}^n l_{ij}\tilde{p}_{2j} + a_{i0}\tilde{p}_{2i}$.

Define $\xi_1 = \mathrm{col}(\xi_{11}, \xi_{12}, \ldots, \xi_{1n})$, $\xi_2 = \mathrm{col}(\xi_{21}, \xi_{22}, \ldots, \xi_{2n})$, $\tilde{p}_1 = \mathrm{col}(\tilde{p}_{11}, \tilde{p}_{12}, \ldots, \tilde{p}_{1n})$, and $\tilde{p}_2 = \mathrm{col}(\tilde{p}_{21}, \tilde{p}_{22}, \ldots, \tilde{p}_{2n})$. Then, we have $\xi_1 = \bar{M}\tilde{p}_1$ and $\xi_2 = \bar{M}\tilde{p}_2$, where $\bar{M} = (L + A_0) \otimes I_m$.

It follows from (19) that the dynamic equations for $\tilde{p}_1$ and $\tilde{p}_2$ are

$$\begin{cases} \dot{\tilde{p}}_1 = \tilde{p}_2 - (\rho_1\theta_1 + \rho_2)\bar{M}\tilde{p}_1 \\ \dot{\tilde{p}}_2 = -(\rho_3\theta_1 + \rho_4)\bar{M}\tilde{p}_2 - X_0 \end{cases} \tag{20}$$

where $X_0 = \ddot{x}_0 \otimes 1_m$ and 1_m denotes an $m \times 1$ vector with all elements being one. The following theorem states the PPT stability of the observer errors $\tilde{p}_1$ and $\tilde{p}_2$.

Theorem 2: Under Assumptions 1–4, consider system (16) with the observer (19) in which the parameters $\rho_k(k = 1, 2, 3, 4)$ are chosen such that $\rho_1 \geq \lambda_{\max}(Q)/\lambda_{\min}(N)$, $\rho_2 \geq (\lambda_{\max}(Q))^2/\lambda_{\min}(N)$, $\rho_3 \geq \lambda_{\max}(Q)/\lambda_{\min}(N)$, and $\rho_4 \geq (\lambda_{\max}(Q))^2/\lambda_{\min}(N)$, where $Q \in R^{n\times n}$ and $N \in R^{n\times n}$ are defined in Theorem 1. The observer errors $\tilde{p}_1$ and $\tilde{p}_2$ will converge to the regions $\|\tilde{p}_1\| \leq \Delta_{p1}$ and $\|\tilde{p}_2\| \leq \Delta_{p2}$ within the predetermined time $2t_{f1}$ and t_{f1}, respectively, where $\Delta_{p2} = \sqrt{X_M^2/(\rho\lambda_{\min}(Q)) + \epsilon_1 V_{p2}(0)/\lambda_{\min}(Q)}$ and $\Delta_{p1} = \sqrt{\Delta_{p2}^2/(\rho\lambda_{\min}(Q)) + \epsilon_1 V_{p1}(t_{f1})/\lambda_{\min}(Q)}$ are some positive constants.

Proof: Consider the following Lyapunov function:

$$V_{p2} = \tilde{p}_2^T \bar{Q} \tilde{p}_2 \tag{21}$$

where $\bar{Q} = Q \otimes I_m$. Calculating the time derivative of V_{p2} and employing (20) yield

$$\begin{aligned}\dot{V}_{p2} &= -(\rho_3\theta_1 + \rho_4)\tilde{p}_2^T\big(\bar{Q}\bar{M} + \bar{M}^T\bar{Q}\big)\tilde{p}_2 - 2\tilde{p}_2^T\bar{Q}X_0 \\ &\leq -(\rho_3\theta_1 + \rho_4)\lambda_{\min}(N)\tilde{p}_2^T\tilde{p}_2 + (\lambda_{\max}(Q))^2\|\tilde{p}_2\|^2 \\ &\quad + \|X_0\|^2 \\ &\leq -\theta_1 V_{p2} - \lambda_{\min}(N)\left(\rho_4 - \frac{(\lambda_{\max}(Q))^2}{\lambda_{\min}(N)}\right)\tilde{p}_2^T\tilde{p}_2 \\ &\quad + X_M^2 \\ &\leq -\theta_1 V_{p2} + X_M^2\end{aligned} \tag{22}$$

where X_M is a positive constant satisfying $\|X_0\| \leq X_M$. By Corollary 1, we can obtain that V_{p2} converges to the region $V_{p2} \leq X_M^2/\rho + \epsilon_1 V_{p2}(0)$ in a predesignated time t_{f1}, and hence the observer error $\tilde{p}_2$ will reach to the region $\|\tilde{p}_2\| \leq \sqrt{X_M^2/(\rho\lambda_{\min}(Q)) + \epsilon_1 V_{p2}(0)/\lambda_{\min}(Q)} = \Delta_{p2}$ in a preset time t_{f1}.

Next, we show the PPT stability of the observer error $\tilde{p}_1$. Consider the Lyapunov function $V_{p1} = \tilde{p}_1^T\bar{Q}\tilde{p}_1$. When $t \geq t_{f1}$, the time derivative of V_{p1} along (20) is

$$\begin{aligned}\dot{V}_{p1} &= -(\rho_1\theta_1 + \rho_2)\tilde{p}_1^T\big(\bar{Q}\bar{M} + \bar{M}^T\bar{Q}\big)\tilde{p}_1 + 2\tilde{p}_1^T\bar{Q}\tilde{p}_2 \\ &= -(\rho_1\theta_1 + \rho_2)\tilde{p}_1^T(N \otimes I_m)\tilde{p}_1 + 2\tilde{p}_1^T\bar{Q}\tilde{p}_2 \\ &\leq -\theta_1 V_{p1} - \rho_2\lambda_{\min}(N)\tilde{p}_1^T\tilde{p}_1 + (\lambda_{\max}(Q))^2\tilde{p}_1^T\tilde{p}_1 + \Delta_{p2}^2 \\ &\leq -\theta_1 V_{p1} + \Delta_{p2}^2\end{aligned} \tag{23}$$

where the following inequalities:

$$\begin{aligned}-\rho_1\theta_1\tilde{p}_1^T(N \otimes I_m)\tilde{p}_1 &\leq -\rho_1\theta_1\lambda_{\min}(N)\tilde{p}_1^T\tilde{p}_1 \leq -\theta_1 V_{p1} \\ 2\tilde{p}_1^T\bar{Q}\tilde{p}_2 &\leq (\lambda_{\max}(Q))^2\tilde{p}_1^T\tilde{p}_1 + \Delta_{p2}^2\end{aligned}$$

have been used. It follows from Corollary 1 that V_{p1} approaches to the region $V_{p1} \leq \Delta_{p2}^2/\rho + \epsilon_1 V_{p1}(t_{f1})$ in a predesignated time t_{f1}; that is, the observer error $\tilde{p}_1$ will converge to the region $\|\tilde{p}_1\| \leq \sqrt{\Delta_{p2}^2/(\rho\lambda_{\min}(Q)) + \epsilon_1 V_{p1}(t_{f1})/\lambda_{\min}(Q)} = \Delta_{p1}$ in a preset time $2t_{f1}$. The proof is complete. ■

B. Design of PPT Extended-State Observers

Let $\hat{x}_{1i}$, $\hat{x}_{2i}$, and $\hat{x}_{3i}$ be the estimates of x_{1i}, x_{2i}, and x_{3i}, respectively, where $x_{3i} = \vartheta_i$ $(i \in \mathcal{N})$. A predefined-time extended-state observer is proposed as

$$\begin{cases}\dot{\hat{x}}_{1i} = \hat{x}_{2i} + \gamma_1\theta_2\tilde{x}_{1i} + 2\frac{\dot{\theta}_2}{\theta_2}\tilde{x}_{1i} \\ \dot{\hat{x}}_{2i} = \hat{x}_{3i} + \hat{f}_i + g_i(x_{1i})u_i + \gamma_2\theta_2^2\tilde{x}_{1i} + \frac{\dot{\theta}_2}{\theta_2}\tilde{x}_{1i} \\ \dot{\hat{x}}_{3i} = \gamma_3\theta_2^3\tilde{x}_{1i}\end{cases} \tag{24}$$

where $\tilde{x}_{1i} = x_{1i} - \hat{x}_{1i}$, $\hat{f}_i = f_i(x_{1i}, \hat{x}_{2i})$, $\gamma_j > 0 (j = 1, 2, 3)$, and $\theta_2 = \gamma_4(\gamma + \dot{\varsigma}_2/\varsigma_2)$. Here, $\varsigma_2 = \varsigma(t, t_{f2}, \epsilon_2)$ with $t_{f2} > 0$ and $\epsilon_2 > 0$, $\gamma_4 \geq \lambda_{\max}(P)$, and $\gamma > 0$, where $P = P^T$ is a positive-definite matrix to be determined as follows.

Defining the observer errors $\tilde{x}_{ji} = x_{ji} - \hat{x}_{ji}$ $(j = 1, 2, 3)$, we have

$$\begin{cases}\dot{\tilde{x}}_{1i} = \tilde{x}_{2i} - \gamma_1\theta_2\tilde{x}_{1i} - 2\frac{\dot{\theta}_2}{\theta_2}\tilde{x}_{1i} \\ \dot{\tilde{x}}_{2i} = \tilde{x}_{3i} + f_i - \hat{f}_i - \gamma_2\theta_2^2\tilde{x}_{1i} - \frac{\dot{\theta}_2}{\theta_2}\tilde{x}_{1i} \\ \dot{\tilde{x}}_{3i} = \dot{\vartheta}_i - \gamma_3\theta_2^3\tilde{x}_{1i}.\end{cases} \tag{25}$$

Considering the variable transformation $\eta_{1i} = \theta_2^2\tilde{x}_{1i}$, $\eta_{2i} = \theta_2\tilde{x}_{2i}$, and $\eta_{3i} = \tilde{x}_{3i}$, we have

$$\begin{cases}\dot{\eta}_{1i} = \theta_2(\eta_{2i} - \gamma_1\eta_{1i}) \\ \dot{\eta}_{2i} = \theta_2(\eta_{3i} - \gamma_2\eta_{1i}) + \theta_2\Big(f_i - \hat{f}_i\Big) \\ \dot{\eta}_{3i} = -\theta_2\gamma_3\eta_{1i} + \dot{\vartheta}_i.\end{cases} \tag{26}$$

Equation (26) can be rewritten as

$$\dot{\eta}_i = \theta_2 H\eta_i + \begin{pmatrix}0 \\ \theta_2(f_i - \hat{f}_i) \\ \dot{\vartheta}_i\end{pmatrix} \tag{27}$$

where $\eta_i = \mathrm{col}(\eta_{1i}, \eta_{2i}, \eta_{3i})$ and $H \in R^{(3m)\times(3m)}$ is defined as

$$H = \begin{bmatrix}-\gamma_1 I_m & I_m & 0 \\ -\gamma_2 I_m & 0 & I_m \\ -\gamma_3 I_m & 0 & 0\end{bmatrix}. \tag{28}$$

If the parameters γ_1, γ_2, and γ_3 are chosen so that matrix H is Hurwitz, then it can be made a conclusion that there exists $P = P^T > 0$ such that $H^T P + PH = -I_{3m}$.

Consider the Lyapunov function

$$V_{oi} = \eta_i^T P\eta_i. \tag{29}$$

The following theorem states the PPT stability of the proposed extended-state observer (24).

Theorem 3: Under Assumptions 1–4, consider system (16) with the extended-state observer (24) in which the parameters γ_1, γ_2, and γ_3 are selected such that the matrix H is Hurwitz, the parameter γ_4 satisfies $\gamma_4 \geq \lambda_{\max}(P)$, and the parameter γ is chosen such that $\gamma_0 = \gamma - 2c\lambda_{\max}(P)/\lambda_{\min}(P) - (\lambda_{\max}(P))^2/\lambda_{\min}(P) > 0$. Then, the observer error $\eta_i (i \in \mathcal{N})$ will converge to the region $\|\eta_i\| \leq \sqrt{\epsilon V_{oi}(0)/\lambda_{\min}(P) + \vartheta_M^2/(\kappa\lambda_{\min}(P))}$ within a predesignated time t_{f2}.

Proof: Taking the time derivative of V_{oi} and applying (27) yield

$$\begin{aligned}\dot{V}_{oi} &= -\gamma_4\left(\gamma + \frac{\dot{\varsigma}_2}{\varsigma_2}\right)\eta_i^T\eta_i + 2\eta_i^T P\begin{pmatrix}0\\ \theta_2(f_i - \hat{f}_i)\\ \dot{\vartheta}_i\end{pmatrix}\\ &\leq -\left(\gamma + \frac{\dot{\varsigma}_2}{\varsigma_2}\right)V_{oi} + 2c\theta_2\lambda_{\max}(P)\|\eta_i\|\|\tilde{x}_{2i}\|\\ &\quad + 2\lambda_{\max}(P)\|\eta_i\|\|\dot{\vartheta}_i\|\\ &\leq -\left(\gamma - \frac{2c\lambda_{\max}(P)}{\lambda_{\min}(P)} - \frac{(\lambda_{\max}(P))^2}{\lambda_{\min}(P)} + \frac{\dot{\varsigma}_2}{\varsigma_2}\right)V_{oi}\\ &\quad + \vartheta_M^2\\ &= -\left(\gamma_0 + \frac{\dot{\varsigma}_2}{\varsigma_2}\right)V_{oi} + \vartheta_M^2\end{aligned} \tag{30}$$

where the following inequalities:

$$\begin{aligned}-\gamma_4\eta_i^T\eta_i &\leq -V_{oi}\\ 2c\theta_2\lambda_{\max}(P)\|\eta_i\|\|\tilde{x}_{2i}\| &\leq 2c\lambda_{\max}(P)\|\eta_i\|^2 \leq \frac{2c\lambda_{\max}(P)}{\lambda_{\min}(P)}V_{oi}\\ 2\lambda_{\max}(P)\|\eta_i\|\|\dot{\vartheta}_i\| &\leq (\lambda_{\max}(P))^2\|\eta_i\|^2 + \vartheta_M^2\\ &\leq \frac{(\lambda_{\max}(P))^2}{\lambda_{\min}(P)}V_{oi} + \vartheta_M^2\end{aligned}$$

have been used. Then, it follows from Corollary 1 that V_{oi} will converge to the region $V_{oi} \leq \epsilon_2 V_{oi}(0) + \vartheta_M^2/\kappa$ within a predefined time t_{f2}, which implies that η_i will converge to the region

$$\|\eta_i\| \leq \sqrt{\frac{\epsilon_2 V_{oi}(0)}{\lambda_{\min}(P)} + \frac{\vartheta_M^2}{\gamma_0\lambda_{\min}(P)}} \tag{31}$$

within a predesignated time t_{f2}. ■

Remark 4: To guarantee the PPT stability of the observer errors $\tilde{p}_1$ and $\tilde{p}_2$, the parameters $\rho_k (k = 1, 2, 3, 4)$ are required to satisfy

$$\begin{aligned}\rho_1 &\geq \frac{\lambda_{\max}(Q)}{\lambda_{\min}(N)}, \rho_2 \geq \frac{(\lambda_{\max}(Q))^2}{\lambda_{\min}(N)}\\ \rho_3 &\geq \frac{\lambda_{\max}(Q)}{\lambda_{\min}(N)}, \rho_4 \geq \frac{(\lambda_{\max}(Q))^2}{\lambda_{\min}(N)}\end{aligned}$$

which implies that global information [i.e., $\lambda_{\max}(Q)$ and $\lambda_{\min}(N)$] may be required. This condition imposes a limitation on the proposed method. To overcome this limitation, the parameters $\rho_k (k = 1, 2, 3, 4)$ can be chosen sufficiently large such that the above conditions hold.

Remark 5: If we choose the observer parameters γ_4 and γ such that $\theta_2 > 1$, we can verify that the observer error $e_{oi} = \text{col}(\tilde{x}_{1i}, \tilde{x}_{2i}, \tilde{x}_{3i})$ approaches to

$$\|e_{oi}\| \leq \|\eta_i\| \leq \sqrt{\frac{\epsilon_2 V_{oi}(0)}{\lambda_{\min}(P)} + \frac{\vartheta_M^2}{\gamma_0\lambda_{\min}(P)}} = \Delta_{oi} \tag{32}$$

within a predefined time t_{f2}, where $i \in \mathcal{N}$.

Remark 6: In this article, we investigate the problem of PPT consensus tracking control of second-order nonlinear MASs with bounded external disturbances for the case when partial-state measurements are available for feedback and the leader's information is accessible only to some of the follower agents. To address this problem, two PPT observers are presented for each follower agent based on the time-varying function technique. The distributed PPT observer (19) is introduced to estimate the leader's states x_0 and $\dot{x}_0$ and the PPT extended-state observer (24) is applied to estimate the system states x_{1i} and x_{2i} as well as the bounded disturbance $x_{3i} = \vartheta_i$. The major difference between the distributed PPT observer (19) and the PPT extended-state observer (24) lies in the fact that the distributed observer for an individual agent requires its neighbors' information while the extended-state observer does not.

C. Design of Distributed PPT Output-Feedback Consensus Tracking Control Law

Define $\hat{e}_{1i} = x_{1i} - p_{1i}$ and $\hat{e}_{2i} = \hat{x}_{2i} - p_{2i} (i \in \mathcal{N})$. The dynamics for $\hat{e}_{1i}$ and $\hat{e}_{2i}$ is

$$\begin{cases}\dot{\hat{e}}_{1i} = x_{2i} - p_{2i} + (\rho_1\theta_1 + \rho_2)\xi_{1i}\\ \qquad = \hat{e}_{2i} + \tilde{x}_{2i} + (\rho_1\theta_1 + \rho_2)\xi_{1i}\\ \dot{\hat{e}}_{2i} = \hat{x}_{3i} + \hat{f}_i + g_i u_i + \varrho_i\end{cases} \tag{33}$$

where $\varrho_i = \gamma_2\theta_2^2\tilde{x}_{1i} + (\dot{\theta}_2/\theta_2)\tilde{x}_{1i} + (\rho_3\theta_1 + \rho_4)\xi_{2i}$.

Design a sliding-mode surface as

$$s_i = \hat{e}_{2i} + \omega_i \tag{34}$$

where $\omega_i \in R^m$ is defined as

$$\omega_i = \frac{k_1 + \theta_3}{2}\hat{e}_{1i} + k_2\tanh\left(\frac{mk_u k_2\hat{e}_{1i}}{\epsilon_0}\right) \tag{35}$$

where $\theta_3 = \dot{\varsigma}_3/\varsigma_3$, $k_1 > 0, k_2 > 0$, and $\epsilon_0 > 0$, and the parameter k_u needs to satisfy $k_u = e^{-(k_u+1)}$, that is, $k_u = 0.2785$ [41]. Here, $\varsigma_3 = \varsigma(t, t_{f3}, \epsilon_3)$ with $t_{f3} > 0$ and $\epsilon_3 > 0$.

The dynamics for $s_i (i \in \mathcal{N})$ is

$$\dot{s}_i = \hat{x}_{3i} + \hat{f}_i + g_i u_i + \dot{\omega}_i + \varrho_i. \tag{36}$$

The distributed control law is chosen as

$$\begin{aligned}u_i &= g_i^{-1}\left(-\frac{k_3 + \theta_3}{2}s_i - \hat{f}_i - \hat{x}_{3i} - \kappa\tanh\left(\frac{mk_u\kappa s_i}{\epsilon_0}\right)\right)\\ &\quad - g_i^{-1}\dot{\omega}_i\end{aligned} \tag{37}$$

where $\kappa > 0$ is a user-defined parameter.

Substituting the control law (37) into (36) yields

$$\dot{s}_i = -\frac{k_3 + \theta_3}{2}s_i - \kappa\tanh\left(\frac{mk_u\kappa s_i}{\epsilon_0}\right) + \varrho_i. \tag{38}$$

Theorem 4: Under Assumptions 1–4, consider system (16) with the distributed PPT observer (19), the PPT extended-state observer (24), and the control law (37). Then, the sliding surface s_i and the tracking error $e_{1i} = x_{1i} - x_0 (i \in \mathcal{N})$ converge to the regions $\|s_i\| \leq \Delta_{si}$ and $\|e_{1i}\| \leq \Delta_{ei}$ within the predesignated time t_{f3} and $t_2 + t_{f3}$, respectively, where the constants $t_2 > 0$, $\Delta_{si} > 0$, and $\Delta_{ei} > 0$ will be defined as follows.

Proof: Consider the Lyapunov function $V_{si} = s_i^T s_i/2 (i \in \mathcal{N})$. Calculating the time derivative of V_{si} and using (38) result in

$$\dot{V}_{si} = -\frac{k_3 + \theta_3}{2}s_i^T s_i + s_i^T\left(\varrho_i - \kappa\tanh\left(\frac{mk_u\kappa s_i}{\epsilon_0}\right)\right). \tag{39}$$

When $t \geq t_1 = \max(2t_{f1}, t_{f2})$, by Theorems 2 and 3, we can obtain that there exists a positive constant ϱ_M such that $\|\varrho_i\| \leq \varrho_M$. The parameter κ is chosen to satisfy $\kappa \geq \varrho_M$.

The hyperbolic tangent function $\tanh(\cdot)$ satisfies the following inequality [41]:

$$0 \leq |z| - z\tanh\left(\frac{z}{\epsilon}\right) \leq k_u\epsilon, \forall z \in R,\ \forall \epsilon > 0. \tag{40}$$

It follows from the inequality (40) that:

$$\begin{aligned} s_i^T&\left(\varrho_i - \kappa\tanh\left(\frac{mk_u\kappa s_i}{\epsilon_0}\right)\right) \\ &\leq \sum_{j=1}^{m}\left(\kappa|s_{ij}| - \kappa s_{ij}\tanh\left(\frac{mk_u\kappa s_{ij}}{\epsilon_0}\right)\right) \\ &\leq \sum_{j=1}^{m}\frac{\epsilon_0}{m} = \epsilon_0 \end{aligned} \tag{41}$$

where the fact that $|\varrho_{ij}| \leq \|\varrho_i\| \leq \kappa$ has been applied. Submitting (41) into (39) yields

$$\dot{V}_{si} \leq -(k_3 + \theta_3)V_{si} + \epsilon_0. \tag{42}$$

By Corollary 1, we can conclude that $V_{si}(i \in \mathcal{N})$ converges to the region $V_{si} \leq \epsilon_3 V_{si}(t_1) + \epsilon_0/k_3$ within a predetermined time t_{f3}, which means that s_i approaches to $\|s_i\| \leq \sqrt{2\epsilon_3 V_{si}(t_1) + \epsilon_0/k_3} = \Delta_{si}$ within a prespecified time $t_1 + t_{f3}$.

Noting that $\hat{e}_{2i} = s_i - \omega_i$, we have

$$\begin{aligned} \dot{\hat{e}}_{1i} &= \hat{e}_{2i} + \tilde{x}_{2i} + (\rho_1\theta_1 + \rho_2)\xi_{1i} \\ &= -\omega_i + s_i + \tilde{x}_{2i} + (\rho_1\theta_1 + \rho_2)\xi_{1i} \\ &= -\frac{k_1 + \theta_3}{2}\hat{e}_{1i} - k_2\tanh\left(\frac{mk_uk_2\hat{e}_{1i}}{\epsilon_0}\right) + h_i \end{aligned} \tag{43}$$

where $h_i = s_i + \tilde{x}_{2i} + (\rho_1\theta_1 + \rho_2)\xi_{1i}$. When $t \geq t_2 = t_1 + t_{f3}$, there exists a positive constant h_M such that $\|h_i\| \leq h_M$. We select the parameter k_2 to meet $k_2 \geq h_M$. It follows from the inequality (40) that:

$$\hat{e}_{1i}^T\left(h_i - k_2\tanh\left(\frac{mk_uk_2\hat{e}_{1i}}{\epsilon_0}\right)\right) \leq \epsilon_0. \tag{44}$$

Consider the Lyapunov function $V_{ei} = \hat{e}_{1i}^T\hat{e}_{1i}/2$. Differentiating V_{ei} with respect to the time t and using (44) lead to

$$\begin{aligned} \dot{V}_{ei} &\leq -\frac{1}{2}(k_1 + \theta_3)\hat{e}_{1i}^T\hat{e}_{1i} + \epsilon_0 \\ &= -(k_1 + \theta_3)V_{ei} + \epsilon_0. \end{aligned} \tag{45}$$

By Corollary 1, it can be concluded that $V_{ei}(i \in \mathcal{N})$ converges to the region $V_{ei} \leq \epsilon_0/k_1 + \epsilon_3 V_{ei}(t_2)$ within a preset time t_{f3} and, thus, $\hat{e}_{1i}$ approaches to the region $\|\hat{e}_{1i}\| \leq \sqrt{2\epsilon_0/k_1 + 2\epsilon_3 V_{ei}(t_2)}$ within a predefined time $t_2 + t_{f3}$. By noting that $e_{1i} = x_{1i} - x_0 = \hat{e}_{1i} + \tilde{p}_{1i}$, we can make a conclusion that the tracking errors e_{1i} converges to the region $\|e_{1i}\| \leq \sqrt{2\epsilon_0/k_1 + 2\epsilon_3 V_{ei}(t_2)} + \Delta_{p1} = \Delta_{ei}$ within a prespecified time $t_2 + t_{f3}$. ∎

Remark 7: Note that when $t > t_2 + t_{f3}$, we can obtain

$$\hat{e}_{2i} = s_i - \omega_i = s_i - \frac{k_1}{2}\hat{e}_{1i} - k_2\tanh\left(\frac{mk_uk_2\hat{e}_{1i}}{\epsilon_0}\right) \tag{46}$$

with $i \in \mathcal{N}$, which implies that $\hat{e}_{2i}$ will converge to the region $\|\hat{e}_{2i}\| \leq \Delta_{si} + k_1\Delta_{ei}/2 + k_2\sqrt{m}$ within a predefined time. By noting that $e_{2i} = x_{2i} - \dot{x}_0 = x_{2i} - \hat{x}_{2i} + \hat{x}_{2i} - p_{2i} + p_{2i} - \dot{x}_0 = \tilde{x}_{2i} + \hat{e}_{2i} + \tilde{p}_{2i}$, it can be obtained that e_{2i} will converge to the region $\|e_{2i}\| \leq \Delta_{si} + k_1\Delta_{ei}/2 + k_2\sqrt{m} + \Delta_{p2} + \Delta_{oi}$ within a predefined time.

Remark 8: It can be obtained from Theorems 2–4 and Remark 4 that the convergent time of the observer errors (i.e., $\tilde{p}_1, \tilde{p}_2$, and e_{oi}) and the tracking errors [i.e., $e_{1i}(i \in \mathcal{N})$] can be predefined by the parameters t_{f1}, t_{f2}, and t_{f3}. That is, the observer errors (i.e., $\tilde{p}_1, \tilde{p}_2$, and e_{oi}) and the tracking errors (i.e., e_{1i}) will converge to the regions $\|\tilde{p}_1\| \leq \Delta_{p1}, \|\tilde{p}_2\| \leq \Delta_{p2}, \|e_{oi}\| \leq \Delta_{oi}$, and $\|e_{1i}\| \leq \Delta_{ei}$ within the predefined time $2t_{f1}, t_{f1}, t_{f2}$, and $\max(2t_{f1}, t_{f2}) + t_{f3}$, respectively, where $\Delta_{p1}, \Delta_{p2}, \Delta_{oi}$, and Δ_{ei} are, respectively, defined by

$$\begin{aligned} \Delta_{p2} &= \sqrt{\frac{X_M^2}{\rho\lambda_{\min}(Q)} + \frac{\epsilon_1 V_{p2}(0)}{\lambda_{\min}(Q)}} \\ \Delta_{p1} &= \sqrt{\frac{\Delta_{p2}^2}{\rho\lambda_{\min}(Q)} + \frac{\epsilon_1 V_{p1}(t_{f1})}{\lambda_{\min}(Q)}} \\ \Delta_{oi} &= \sqrt{\frac{\epsilon_2 V_{oi}(0)}{\lambda_{\min}(P)} + \frac{\vartheta_M^2}{\gamma_0\lambda_{\min}(P)}} \\ \Delta_{ei} &= \sqrt{\frac{2\epsilon_0}{k_1} + 2\epsilon_3 V_{ei}(t_2)} + \Delta_{p1} \end{aligned}$$

which implies the parameters $\rho, \gamma(\gamma_0 = \gamma - 2c\lambda_{\max}(P)/\lambda_{\min}(P) - (\lambda_{\max}(P))^2/\lambda_{\min}(P)), k_1, \epsilon_0, \epsilon_1, \epsilon_2$, and ϵ_3 can determine the final accuracy of $\tilde{p}_1, \tilde{p}_2, e_{oi}$, and e_{1i}; that is, the smaller the errors $\tilde{p}_1, \tilde{p}_2, e_{oi}$, and e_{1i}, the larger the parameters ρ, γ, and k_1 and the smaller the parameters $\epsilon_1, \epsilon_2, \epsilon_3$, and ϵ_0 are required.

Remark 9: In [23], a distributed fixed-time velocity-free consensus tracking control law was proposed. Since there are some terms with a power greater than one in the designed controller, a large magnitude of initial control inputs might be required to achieve fixed-time stability. However, the proposed predefined-time control scheme has an advantage in reducing the magnitude of the applied control inputs.

Remark 10: In practice, a discontinuous control law may cause chattering. However, chattering is in general not desirable in practical situations since it may excite high-frequency unmodeled system dynamics. To avoid the chattering caused by the use of the discontinuous sign function, the smooth hyperbolic tangent function is used in the sliding-mode surface (34) and the proposed control law (37). The hyperbolic tangent function can be considered as a smooth approximation of the discontinuous sign function.

Remark 11: In [31], the terminal sliding-mode control approach was employed to design a predefined-time consensus tracking control scheme for double-integrator MASs. While the communication graph is undirected and full-state measurements are required in [31], the directed communication topology is considered and full-state measurements are not necessary in this article.

Remark 12: In [35], a distributed consensus tracking controller was proposed for second-order MASs with bounded disturbances based on the time-varying function technique. There are five main differences between the present paper and [35]. First, double-integrator dynamics is investigated in [35], but second-order nonlinear systems are explored in the present paper. Second, the control scheme presented in [35] is discontinuous while the control law proposed in this article is continuous. Third, the assumption of full-state measurements is required in [35] but such an assumption is relaxed in this article. Fourth, a communication loop problem may occur in implementing the controller designed in [35] while there is no loop problem in implementing the proposed controller. Fifth, the communication graph considered in [35] is signed directed while the fixed directed communication topology is taken into account in this article. Nonetheless, how to design a distributed PPT output-feedback consensus control scheme for second-order nonlinear MASs under signed directed graphs deserves further investigation.

Remark 13: One disadvantage of the proposed approach is the requirement of continuous communication among neighboring agents to implement the proposed controller. However, such an ideal communication condition might not be guaranteed in practical applications because of the communication bandwidth and limited energy power. The event-triggered control was employed to design a predefined-time consensus controller for single-integrator MASs in [32]. Thus, one interesting future research is to analyze and design a distributed event-triggered PPT output-feedback consensus controller for MASs.

Remark 14: The PPT output-feedback consensus tracking control problem can be divided into three subproblems: 1) the distributed PPT state observer design problem; 2) the PPT extended-state observer design problem; and 3) the PPT controller design problem. Distributed predefined-time observers were proposed in [31] and [37]. However, the communication topology considered in [31] was undirected and the unbounded time-varying function was used in [37]. Moreover, the design approaches for a predefined-time controller may include the sliding-mode control method and the time-vary function technique. However, some of the existing sliding-mode control-based and the time-varying function-based predefined-time control schemes may suffer from a singularity problem due to the presence of some terms with a negative power in the sliding-mode surface [26], [27], [28] and unbounded control gains in the designed controllers [32], [33], [36], [37]. Hence, the main technical difficulties behind the PPT output-feedback consensus tracking controller design arise from the PPT observer design problem and the control singularity avoidance problem.

IV. Simulation Results

In this section, the performance of the designed control scheme (37) is illustrated through numerical simulations for the situation when there are a group of six follower agents and one virtual leader is taken into consideration. The matrices A and A_0 are taken, respectively, as

$$A = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \; A_0 = \mathrm{diag}(1, 0, 0, 0, 0, 1).$$

The nonlinear function $f_i(x_{1i}, x_{2i})$ is $f_i = 2\sin(x_{1i}) - 0.4x_{2i}/i$ $(i = 1, 2, \ldots, 6)$ and $g_i = I_2$, where $x_{1i}, x_{2i} \in R^2$. As shown in Remark 6, the smaller the observer errors $\tilde{p}_1, \tilde{p}_2$, and e_{oi} and the tracking errors $e_{1i}(i = 1, 2, \ldots, 6)$, the larger the parameters ρ, γ, and k_1 and the smaller the parameters $\epsilon_1, \epsilon_2, \epsilon_3$, and ϵ_0 are required. Thus, the parameters $\rho, \gamma, \epsilon_0, \epsilon_1, \epsilon_2$, and ϵ_3 are chosen as $\rho = \gamma = 100, \epsilon_0 = 0.001$, and $\epsilon_k = 0.0001 (k = 1, 2, 3)$, respectively. Other observer and controller parameters are set as $\rho_1 = 2, \rho_2 = 5, \rho_3 = 2, \rho_4 = 5, \gamma_k = 2 (k = 1, 2, 3, 4), k_1 = k_2 = 0.5, k_3 = 1.5$, and $t_{f1} = t_{f2} = 2$ s. The disturbances ϑ_i are supposed to be $\vartheta_i = 0.1\mathrm{col}(\cos(it), 0.5\sin(it))$. The reference state is $x_0 = \mathrm{col}(\cos(t/5), \sin(t/5))$. The initial states of the observers (19) and (24) are $p_{1i}(0) = p_{2i}(0) = 0$, $\hat{x}_{1i}(0) = x_{1i}(0)$, and $\hat{x}_{2i}(0) = \hat{x}_{3i}(0) = 0$.

To study the performance of the designed control scheme, two tracking error metrics (i.e., TEM_1 and TEM_2) are introduced and they are, respectively, defined as

$$\mathrm{TEM}_1 = \sqrt{\sum_{i=1}^{6} \|e_{1i}\|^2}, \; \mathrm{TEM}_2 = \sqrt{\sum_{i=1}^{6} \|e_{2i}\|^2}$$

where $e_{2i} = x_{2i} - \dot{x}_0$. Three cases are taken into consideration in the simulation.

Case A: In this case, the parameter t_{f3} is set as $t_{f3} = 20$ s, and initial states are $x_{11}(0) = \mathrm{col}(1, 2)$, $x_{12}(0) = \mathrm{col}(2, -1)$, $x_{13}(0) = \mathrm{col}(-1, 1)$, $x_{14}(0) = \mathrm{col}(-1.5, 1.5)$, $x_{15}(0) = \mathrm{col}(1.5, -2)$, $x_{16}(0) = \mathrm{col}(-2, -1.5)$, and $x_{2i}(0) = 0$. With the control law (37), the system states $x_{1i}(i = 1, 2, \ldots, 6)$ are presented in Fig. 1. The plots for the error metrics TEM_1 and TEM_2 are given in Fig. 2. It can be observed from Fig. 2 that TEM_1 and TEM_2 converge to $\mathrm{TEM}_1 \leq 4.8 \times 10^{-5}$ and $\mathrm{TEM}_2 \leq 2 \times 10^{-5}$ within $t = t_{f1} + t_{f3} = 22$ s. The bounded and continuous applied control inputs are shown in Fig. 3. The results indicate that the states of all agents can achieve an agreement on the leader's state even when only partial-state measurements are available for feedback and a part of the team members can have access to information from the leader.

Case B: The effect of the predefined time t_{f3} on the performance of the present control law is examined. Four performance metrics, that is, $E_{1\max}, E_{2\max}, u_{\max}$, and E, are introduced for the comparison, where $E_{1\max}$ and $E_{2\max}$ denote the maximum values of $\mathrm{TEM}_1(t)$ and $\mathrm{TEM}_2(t)$ when $t \geq t_{f1} + t_{f3}$, respectively, $u_{\max} = \max(|u_{ik}(t)|), i = 1, 2, \ldots, 6, k = 1, 2$, and E is employed to measure the total consumed energy given by

$$E = \int_{t=0}^{T} \sum_{i=1}^{6} \|u_i(t)\|^2 dt \tag{47}$$

with $T = 40$ s. In this case, t_{f3} is selected as 5, 10, and 30 s, respectively. The results are given in Table I. It can be seen from Table I that with the increase of t_{f3}, the magnitude of the applied control inputs (i.e., $u_{\max}$) and the consumed energy become smaller. When $t_{f3} = 5$ s, $E_{1\max} = 8 \times 10^{-3}$ and $E_{2\max} = 1.2 \times 10^{-1}$ which are larger than those for the cases when $t_{f3} = 10$ s and $t_{f3} = 30$ s. Note that when we calculate $E_{1\max}$ and $E_{2\max}$, the settling time is set as $t = t_{f1} + t_{f3}$ which is less than the preset time claimed in Theorem 4 (i.e.,

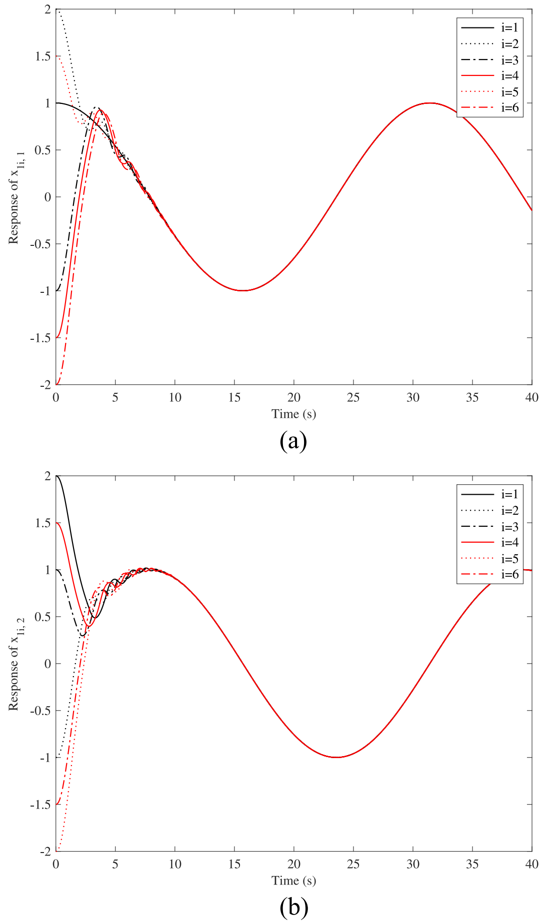

Fig. 1. Response of $x_{1i}(i = 1, 2, \ldots, 6)$. (a) $x_{1i,1}(i = 1, 2, \ldots, 6)$. (b) $x_{1i,2}(i = 1, 2, \ldots, 6)$.

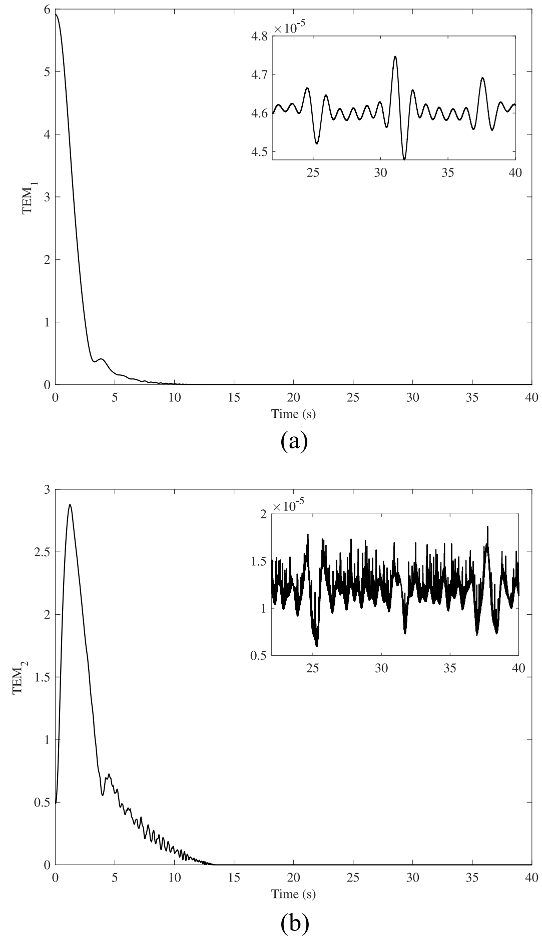

Fig. 2. Response of (a) TEM_1 and (b) TEM_2.

TABLE I
EFFECT OF THE PREDEFINED TIME t_{f3} ON THE PERFORMANCE OF THE PROPOSED CONTROLLER

t_{f3}(s)	$E_{1\,\max}$	$E_{2\,\max}$	$u_{\max}$	E
5	8.0×10^{-3}	1.2×10^{-1}	17.00	1109.6
10	4.8×10^{-5}	2.0×10^{-5}	4.48	1022.0
30	4.7×10^{-5}	2.0×10^{-5}	3.09	1018.2

$t = t_2 + t_{f3} = t_1 + 2t_{f3}$). However, when $t \geq t_1 + t_{f3} = 10$ s, the maximum values of $\text{TEM}_1(t)$ and $\text{TEM}_2(t)$ by the proposed controller with $t_{f3} = 5$ s are 4.95×10^{-5} and 1.0×10^{-3}, respectively.

Case C: The performance of the developed distributed predefined-time output-feedback consensus controller is compared with the distributed velocity-free fixed-time controller presented in [23]. In this example, the parameter t_{f3} is set as $t_{f3} = 20$ s and initial conditions are considered as $x_{11}(0) = i_c\text{col}(1, 2), x_{12}(0) = i_c\text{col}(2, -1), x_{13}(0) = i_c\text{col}(-1, 1), x_{14}(0) = i_c\text{col}(-1.5, 1.5), x_{15}(0) = i_c\text{col}(1.5, -2)$, and $x_{16}(0) = i_c\text{col}(-2, -1.5)$, where i_c is some positive constant. In this example, i_c is chosen as 1, 10, and 100. The results are given in Table II. It can be concluded from Table II that with the increase of c_a (i.e., the increase of initial states and, thus, the increase of initial consensus tracking errors), the consensus tracking performance remains unchanged for the proposed controller ($E_{1\,\max} = 4.8 \times 10^{-5}$ and $E_{2\,\max} = 2.0 \times 10^{-5}$ for $i_c = 1, 10$, and 100) and the fixed-time controller ($E_{1\,\max} = 2.6 \times 10^{-3}$ and $E_{2\,\max} = 8.0 \times 10^{-4}$ for $c_a = 1, 10$, and 100); however, the magnitude of the applied control inputs and the consumed energy for the present controller ($u_{\max} = 10.31$ and $E = 1.32 \times 10^3$ for $i_c = 10$, $u_{\max} = 87.35$ and $E = 2.63 \times 10^4$ for $i_c = 100$) are much smaller than those for the fixed-time controller ($u_{\max} = 37.65$ and $E = 4.52 \times 10^3$ for $i_c = 10$, $u_{\max} = 846.22$ and $E = 1.35 \times 10^6$ for $i_c = 100$). The results indicate that the developed control law can yield good consensus tracking control performance with the requirement of much less energy and a much smaller magnitude of the applied control inputs than the fixed-time controller.

V. CONCLUSION

A new predefined-time output-feedback consensus tracking control scheme has been developed for MASs with the help of the smooth time-dependent scaling function, the distributed state observer, the extended-state observer, and the sliding-mode control. The proposed controller is continuous and nonsingular, and guarantees that the consensus tracking errors converge to a neighborhood of zero within a predesignated time, independent of initial conditions. Illustrative examples were presented to show that the proposed protocol

TABLE II
PERFORMANCE COMPARISON BETWEEN THE PROPOSED CONTROLLER (37) AND THE FIXED-TIME CONTROLLER IN [23]

	$E_{1\max}$	$E_{2\max}$	$u_{\max}$	E
Proposed controller ($i_c = 1$)	4.8×10^{-5}	2.0×10^{-5}	3.10	1.02×10^{3}
Fixed-time controller ($i_c = 1$)	2.6×10^{-3}	8.0×10^{-4}	3.50	1.00×10^{3}
Proposed controller ($i_c = 10$)	4.8×10^{-5}	2.0×10^{-5}	10.31	1.32×10^{3}
Fixed-time controller ($i_c = 10$)	2.6×10^{-3}	8.0×10^{-4}	37.65	4.52×10^{3}
Proposed controller ($i_c = 100$)	4.8×10^{-5}	2.0×10^{-5}	87.35	2.63×10^{4}
Fixed-time controller ($i_c = 100$)	2.6×10^{-3}	8.0×10^{-4}	846.22	1.35×10^{6}

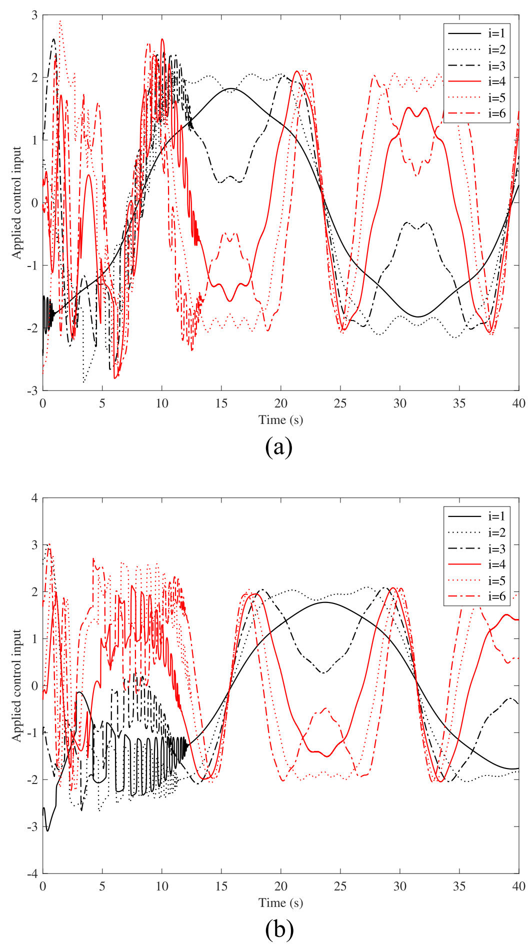

Fig. 3. Applied control input $u_i (i = 1, 2, \ldots, 6)$. (a) $u_{i,1} (i = 1, 2, \ldots, 6)$. (b) $u_{i,2} (i = 1, 2, \ldots, 6)$.

has superiority in saving energy and reducing the magnitude of the applied control inputs.

APPENDIX

The proof for the three properties of the function $\varsigma(t, t_f, \epsilon)$ defined in (6) is given as follows.

Proof of Property 1: When $t = 0$, $\varsigma_0(t) = 0$ and $\varsigma_0(t_f - t) = e^{-1/t_f}$. Thus, we have

$$\varsigma(t, t_f, \epsilon) = \frac{e^{-1/t_f}}{(1+\epsilon)e^{-1/t_f}} = \frac{1}{1+\epsilon}. \tag{48}$$

When $t \geq t_f$, $\varsigma_0(t) = e^{-1/t}$ and $\varsigma_0(t_f - t) = 0$. Hence, we obtain

$$\varsigma(t, t_f, \epsilon) = \frac{e^{-1/t}}{\epsilon e^{-1/t}} = \frac{1}{\epsilon}. \tag{49}$$

Proof of Property 2: By noting that

$$\varsigma(t, t_f, \epsilon) = \frac{1}{\varsigma_a(t, t_f) + \epsilon} = \frac{\varsigma_0(t_f - t) + \varsigma_0(t)}{(1+\epsilon)\varsigma_0(t_f - t) + \epsilon\varsigma_0(t)} \tag{50}$$

to prove that $\varsigma(t, t_f, \epsilon)$ is C^∞ on $(0, +\infty)$, it is only required to show that the function

$$\varsigma_0(x) = \begin{cases} 0, & (-\infty, 0] \\ e^{-1/x}, & (0, +\infty) \end{cases} \tag{51}$$

is C^∞ on $(-\infty, +\infty)$; that is, it is requested to prove that $\varsigma_0^{(n)}(x)$ exists and is continuous on $(-\infty, +\infty)$ for any integer $n > 0$, where $\varsigma_0^{(n)}(x) = d^n\varsigma_0(x)/dx^n$. When $x < 0$, we have $\varsigma_0^{(n)}(x) = 0$ for any $n > 0$. When $x > 0$, it can be obtained that $\varsigma_0'(x) = e^{-1/x}/x^2$ for $n = 1$, and by the mathematical induction method, it can be verified that

$$\varsigma_0^{(n)}(x) = e^{-\frac{1}{x}}\left(\frac{1}{x^{2n}} + \sum_{j=1}^{n-1} a_j \frac{1}{x^{2n-j}}\right) \tag{52}$$

for any $n \geq 2$, where $a_j (j = 1, 2, \ldots, n-1)$ are some constants. This indicates that $\varsigma_0^{(n)}(x)$ exists and is continuous on $(-\infty, 0)$ and $(0, +\infty)$, respectively, for any integer $n > 0$. Moreover, we can verify that $\lim_{x\to 0+} e^{-1/x}/x^2 = 0$ and for any $n \geq 2$

$$\begin{aligned} &\lim_{x\to 0+} e^{-\frac{1}{x}}\left(\frac{1}{x^{2n}} + \sum_{j=1}^{n-1} a_j \frac{1}{x^{2n-j}}\right) \\ &= \lim_{y\to +\infty} \frac{\left(y^{2n} + \sum_{j=1}^{n-1} a_j y^{2n-j}\right)}{e^y} = 0 \end{aligned} \tag{53}$$

where L'Hopital's rule has been repeatedly applied $2n$ times. Therefore, to prove that $\varsigma_0^{(n)}(x)$ exists and is continuous at $x = 0$ for any $n > 0$, we have to show that $\lim_{x\to 0-} \varsigma_0^{(n)}(x) = \lim_{x\to 0+} \varsigma_0^{(n)}(x) = 0$, which will be proven by using the mathematical induction method.

Step 1: When $n = 1$, it is easy to verify that $\varsigma_0(x)$ is continuous at $x = 0$. Furthermore

$$\lim_{x\to 0-} \varsigma_0'(x) = \lim_{x\to 0-} \frac{0}{x} = 0 \tag{54}$$

$$\lim_{x\to 0+} \varsigma_0'(x) = \lim_{x\to 0+} \frac{e^{-1/x}}{x} = 0. \tag{55}$$

Hence, $\lim_{x\to 0-} \varsigma_0'(x) = \lim_{x\to 0+} \varsigma_0'(x) = 0$.

Step 2: When $n = k (k \geq 2)$, assume that $\lim_{x\to 0-} \varsigma_0^{(k)}(x) = \lim_{x\to 0+} \varsigma_0^{(k)}(x) = 0$. Next, we need to show that $\lim_{x\to 0-} \varsigma_0^{(k+1)}(x) = \lim_{x\to 0+} \varsigma_0^{(k+1)}(x) = 0$ when $n = k + 1$.

It is clear that $\lim_{x\to 0-} \varsigma_0^{(k+1)}(x) = \lim_{x\to 0-}(0/x) = 0$. Furthermore

$$\begin{aligned}\lim_{x\to 0+} \varsigma_0^{(k+1)}(x) &= \lim_{x\to 0+} \frac{e^{-\frac{1}{x}}\left(\frac{1}{x^{2k}} + \sum_{j=1}^{k-1} a_j \frac{1}{x^{2k-j}}\right)}{x} \\ &= \lim_{x\to 0+} \frac{\left(\frac{1}{x^{2k+1}} + \sum_{j=1}^{k-1} a_j \frac{1}{x^{2k-j+1}}\right)}{e^{\frac{1}{x}}} \\ &= \lim_{y\to +\infty} \frac{y^{2k+1} + \sum_{j=1}^{k-1} a_j y^{2k-j+1}}{e^y} = 0 \quad (56)\end{aligned}$$

where L'Hopital's rule has been repeatedly used $2k+1$ times. Thus, $\lim_{x\to 0-} \varsigma_0^{(k+1)}(x) = \lim_{x\to 0+} \varsigma_0^{(k+1)}(x) = 0$.

By steps 1 and 2, it can be concluded that the function $\varsigma_0(x)$ is C^∞ on $(-\infty, +\infty)$, which, in turn, implies that $\varsigma(t, t_f, \epsilon)$ is C^∞ on $(0, +\infty)$.

Proof of Property 3: By the proof of Property 2), we can obtain that

$$\varsigma_0'(x) = \begin{cases} 0, & (-\infty, 0] \\ \frac{1}{x^2} e^{-1/x}, & (0, +\infty). \end{cases} \quad (57)$$

When $t \in (0, t_f)$, the time derivative of $\varsigma(t, t_f, \epsilon)$ is

$$\varsigma'(t, t_f, \epsilon) = \frac{\left(\frac{1}{(t_f-t)^2} + \frac{1}{t^2}\right) e^{-\frac{t_f}{t(t_f-t)}}}{\left[(1+\epsilon) e^{-\frac{1}{t_f-t}} + \epsilon e^{-\frac{1}{t}}\right]^2} > 0 \quad (58)$$

which implies that $\varsigma(t, t_f, \epsilon)$ is monotonically increasing on $(0, t_f)$.

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